

EEE 419/591

Fall 2026

Homework 1

Solve the following three problems. Put them all into a single code named hw1.py and submit it to the designated link on Canvas. If you upload a second attempt to Canvas and it renames your file, ignore the renaming.

In this HW, only limited AI is allowed. Examples of allowed prompts on this HW:

- What is a greatest common divisor of two numbers?
- How do I write an if statement in python? Give me an example
- How do I check if a number is integer in python?
- I got this error message. How do I interpret it?

Examples of prohibited prompts on this HW:

- Write me a code that calculates the greatest common divisor of a list of numbers in python
- I ran this code. [PASTE CODE HERE]. I got this error message. [PASTE THE ERROR MESSAGE HERE]. How do I fix my code?

Do not forget to cite your AI chat if you used any. Check the Syllabus document for details.

Problem 1: Least Common Multiple

Write a code that takes a list of numbers and calculates the Least Common Multiple (LCM) of them all. Do not use any built-in “least common multiple” or “lcm” functions from any Python packages.

Your code should expect a list of numbers as an input on the form $[x_1, x_2, x_3, \dots, x_N]$ where x_i is the i^{th} number that could be any real number. Your code should print “The LCM is:” followed by the least common multiple of the list passed to it by the user. If one of the numbers is negative or zero, your code should display an error message, ignore that number, and find the LCM of the remaining positive numbers. Keep in mind that a number formatted as float (e.g., 12.0) can still be a valid positive integer.

An example of the output of your code is as follows:

Input a list of numbers: [-4, 12, 18.0, 0, 24]

Error: Ignoring -4, number must be positive

Error: Ignoring 0, number must be positive

The LCM of the valid integers in this list is: 72

Hint: start writing a code that works on a list of 2 numbers.

Problem 2: Perfect Square Checker

Write a code that takes a list of integers and outputs, for each number, whether it is a perfect square (a whole number that is the square of an integer, such as 1, 4, 9, 16, 25). Do not use any built-in “perfect square” checker or square-root functions from external Python packages (e.g., `math.isqrt`, or `np.sqrt`).

Your code should expect a single list of numbers (possibly not integers) as an input and print if each number is a perfect square. If one of the numbers is negative or not an integer, your code should skip that number and let the user know it is invalid. Keep in mind that a number formatted as float (e.g., 16.0) does not mean it is not an integer.

Consider the following example:

Input a list of numbers: [49.0, 17, -9, 81.4, 36]

49 is a perfect square

17 is NOT a perfect square

-9 is negative. Skipping ...

81.4 is NOT an integer. Skipping ...

36 is a perfect square

Hint: A number N is a perfect square if there exists an integer k such that $k^2 = N$.

Problem 3: Pi Approximation via the Wallis Product

Write a code that calculates an approximation of π using the Wallis Product formula for a specific accuracy or number of terms. Your code should take a single integer as input representing the number of product terms to compute, and output the resulting approximate value of π rounded to 6 decimal places, along with the number of terms used.

Do not use the built-in `math.pi` or `np.pi` constants or any special mathematical functions from external Python packages.

Recall that the Wallis Product for approximating π is given by:

$$\pi = 2 \times \prod_{n=1}^{\infty} \left(\frac{4n^2}{4n^2 - 1} \right)$$

Since we cannot calculate an infinite product, your code will compute a finite product based on the user's input N . Your code should output the result formatted as shown in the example below:

Input the number of terms: 5000

Pi approx = 3.141436 based on 5,000 terms

MANDATORY – Do not forget to add your mandatory citations as indicated on page 4 of the syllabus document. If you have not used any AI/references in your document, you still need to declare that as indicated in the syllabus document. This has 20% weight according to the rubric in the syllabus.